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Projecting College Football Player Performance in the age of NIL and the Transfer Portal

Introduction

In the NIL era of college sports, the transfer portal has become more important than ever when it comes to roster building. Teams such as Curt Cignetti's Indiana have been able to completely flip their rosters around in an offseason or two by pursuing transfers, with differing levels of success. Cignetti's Indiana team this past season contained 33 transfers brought in since he was hired, with 14 of 22 starters on offense and defense coming from the portal. Traditional team talent metrics such as high school recruiting rankings have become less and less relevant, as Indiana was able to go 16-0 and win the national championship without employing a single five-star recruit on offense. Programs increasingly rely on transfers to fill immediate needs, but the industry lacks an objective, data-driven framework for predicting which transfer players will succeed at their new schools. Similar to free agency in professional sports, teams and coaches alike need a way to evaluate players who enter the transfer portal both in terms of player evaluation and team fit. This project will aim to quantify the "value" of any player entering the transfer portal. The framework proposed is position-agnostic, allowing coaches and player personnel decision-makers alike to evaluate the trade-offs of bringing in different numbers of players (e.g. is it better to splurge NIL dollars on a high-profile quarterback, or is the team better off splitting that money among his offensive line).

Literature Review

Current transfer portal research in college football is quite uncommon, with the era of NIL and constant transfers just beginning. Even less publicly available at the college football level is the concept of “one-number” metrics to measure player performance or production. The first phase of this project requires carefully defining a target variable to measure player success or impact on wins in a given season. Pro Football Focus (PFF) maintains a proprietary metric, called Wins Above Average (WAA), to attempt to create a metric similar to Wins Above Replacement in baseball but at the collegiate football level. PFF utilizes their internal player grades to measure how “good” a player was at different aspects of the sport, then maps these grades to wins using some form of regression not mentioned in their writeup. From there, they simulate the season both with and without the player in question to estimate the “expected wins” added by said player. They also provide WAA Projections, where they estimate a given player’s impact in the years to come. PFF doesn’t detail the exact methodology for their college football metric or projections, so this study provides an opportunity to open-source a similar metric subject to academic review as opposed to under full confidentiality of a third-party company. It is worth noting that the NFL equivalent of this model, titled PFF NFL WAR, is detailed in [this paper](#). They describe more of the methodology, utilizing an adjusted Massey matrix weighted by PFF’s player grades, the individual season of play, and the player’s position.

ESPN maintains another version of a one-number success metric for college football players, called [Playstation Player Impact Ratings](#). The methodology described is a form of Regularized Adjusted Plus Minus (RAPM), similar to what we often see in basketball. Since football scoring events are far less common, ESPN utilizes EPA (Estimated Points Added) as the response variable as opposed to actual points. They introduce a Bayesian hierarchical approach

to estimate the value of each player on the field, adjusted for the other 21 players on the field on any given snap. ESPN also adds in adjustments for run plays versus pass plays, as well as including position within the Bayesian hierarchical structure to allow for individualized variance estimates for each position. ESPN keeps the exact nuances of the model hidden, and does not seem to publish the full results of the model anywhere publicly available. This research provides an opportunity to replicate a similar structure for player evaluation with more transparency of methodology and results. ESPN's (limited) results described are only for the 2018 season (before the NIL/transfer portal era), and so this research can determine whether the methods used are still applicable to the new era of college football. Since transfer portal research in football is rare, we can borrow ideas from other sports such as basketball. One such example comes from Evan Miya, who publishes transfer portal rankings in college basketball based on his flagship metric, [Bayesian Performance Rating \(BPR\)](#). Miya's BPR stat incorporates a multi-year RAPM model, used to determine weights for a box-plus minus model based solely off of counting box score stats. He then utilizes his BPM stat as the prior mean for his RAPM model to come up with his BPR ratings for each player. An important thing to note for this research is how Miya builds his preseason projections, utilizing the player's prior years BPM and BPR in addition to recruiting rankings, transfer status, and more. It will be helpful to incorporate similar methods in this college football research, as adding in recruiting rankings, counting-stats data, and some sort of adjusted plus-minus model will likely yield the best results in predicting player performance in the next season.

A more common form of public college football "projections" comes in the form of NFL draft models. These models attempt to predict NFL player success based on college statistics and other data, not dissimilar to this research's goal of predicting college football success based on

prior years of college statistics and other data. One example of such a draft model is Cooper Klaus's from 2024. One note to keep in mind from this article for the college football research is that Klaus chooses to include public big board/scouting data as features in the model. One takeaway from this is that it could be helpful to include prior recruiting rankings from websites such as On3 and 247Sports as features in the model for projecting college football player performance. To add on from there, Klaus determines which metrics correlated most from college to the NFL and incorporated those as features in the model as well. This research can take a similar approach, identifying which metrics correlate the most from one season to the next of college football and including those in the player projection model.

By incorporating aspects from these different studies and models, the research will build off of prior work and provide a robust method for evaluating college football players.

Methods and Analysis

Data Description

For the purposes of modeling individual player value in the transfer portal, I chose to use PFF WAA as a one-number proxy for player performance. Considering the transfer portal regulations allowing college football players to transfer schools without needing to sit out a year were adopted in 2021 (as well as the shortened COVID-19 season in 2020 leading to a far smaller sample size), I chose to limit my modeling focus from the 2021-22 college football season to the present day. The dataset includes any player in college football who had a PFF WAA estimate for any consecutive seasons between 2021 and 2026 (so a player who played in 2021, 2022, 2023 would have two separate observations in our data). I expand the analysis to all players in this timeframe as opposed to just transferring players to allow the model to learn from a greater sample size of players. PFF WAA estimates are downloaded directly from the PFF

Ultimate website, which contains the player's name, team, position, number of seasons previously played in college football, number of snaps, and WAA. The PFF Ultimate website also contains detailed athleticism metrics on the season-level derived by Pro Football Focus's internal computer vision algorithms. These metrics include, but are not limited to the following: maximum player speed over the season, total distance covered on-the-field, average and maximum speed from 0-10 yards and 10-20 yards, acceleration, deceleration, and get-off speed from 0-5 yards, 5-10 yards, 10-15 yards, and 15-20 yards.

In addition to the plentiful PFF Ultimate data, I also include data open-sourced by the CFBFastR package in RStudio. I obtain height and weight using the `cfbd_team_roster()` function, as season-level box score statistics (pass yards, rush yards, touchdowns, interceptions, tackles, etc.) using the `cfbd_stats_season_player()` function. I also obtain recruiting-level data such as ranking, number of stars, and 247Sports rating using the `cfbd_recruiting_player()` function for high school recruiting data and the `cfbd_recruiting_transfer_portal()` function for transfer portal recruiting data. One challenge that comes up when combining both the PFF Ultimate and CFBFastR data into a unified modeling structure is the lack of matching player and team IDs. As such, the first step in the project was creating unified dataframes with every FBS player and team's PFF Ultimate and CFBFastR ID for matching.

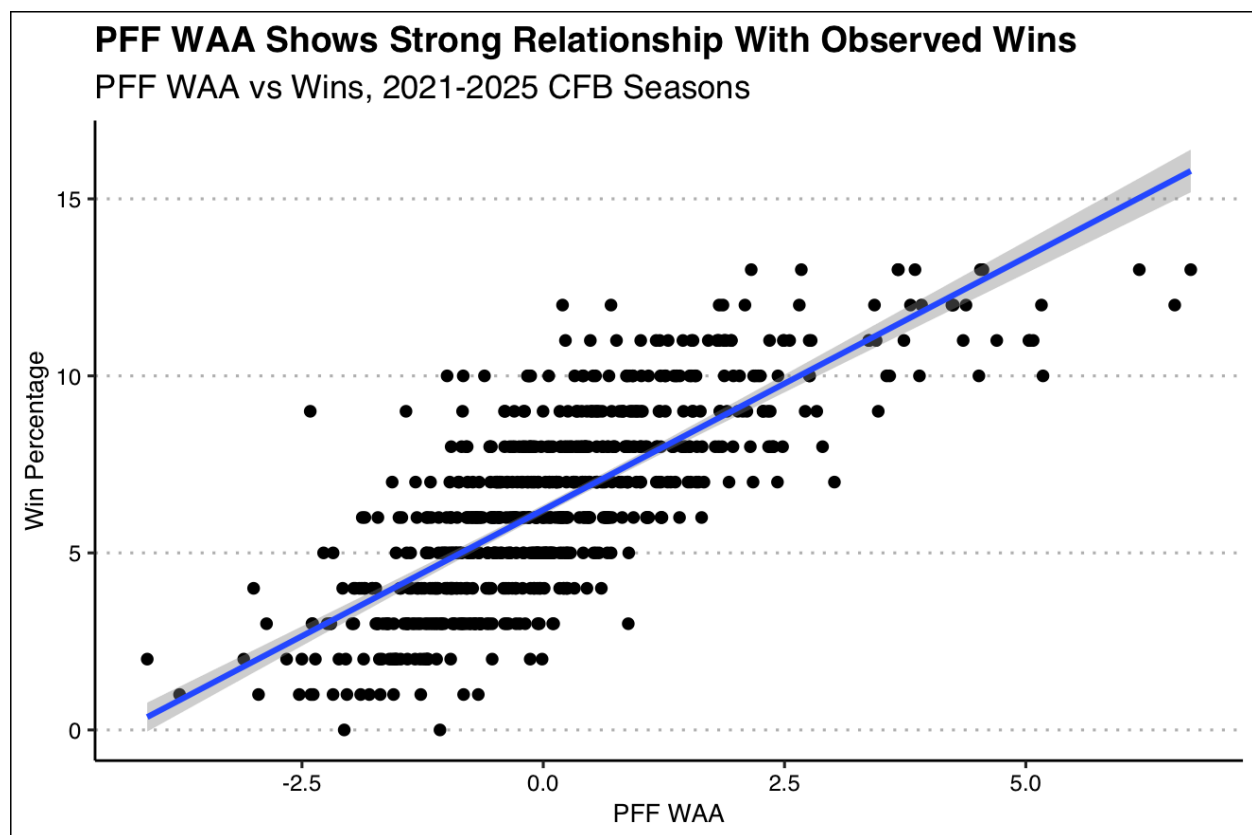
Handling Missing Data

There was a non-insignificant amount of missing data with recruiting rankings, as certain players were never ranked by 247Sports. These represent the unheralded recruits who were not on a national radar before entering the college football ranks. To handle this, all players with missing high school recruiting ranks were tagged as "0-star" recruits, ranked 9999th in their class, with a 0 rating from 247. Imputing the transfer portal rankings data for players with none required a

more careful approach. Players who entered the transfer portal, but were not ranked by 247Sports were once again tagged as “0-star” recruits (as they were not seen as “good enough” to be evaluated by 247). However, players who did not enter the transfer portal were labeled as 9999-star recruits, so the model is able to differentiate between players who entered and did not enter the portal while still incorporating the valuable information gleaned from 247 transfer portal rankings.

Exploring PFF Wins Above Average’s Relationship with Observed Wins

Before committing to PFF WAA as a one-number proxy for player impact, I first chose to explore how the metric related to actual wins at the CFB level. Below we see a plot demonstrating this relationship:



Among all seasons in our data (2021-2025), we see a fairly strong positive correlation between team Wins Above Average and overall team performance. A simple linear regression of wins

regressed on PFF WAA yields a powerful 0.5942 Adjusted R-Squared, suggesting that about ~60% of overall differences between team performance can be explained by the PFF WAA metric.

As mentioned earlier, PFF also maintains their own internal WAA projection system, for predicting WAA in the next year. However, we see a much weaker relationship between projected WAA and wins, with just a 0.2421 Adjusted R-Squared. As such, this project attempts to improve on the WAA projections given by PFF in hope of finding a clearer relationship with game outcomes in the next season.

Projections Methodology

After finishing all the data acquisition and cleaning, we are left with about 24,000 observations of player-seasons in the previously mentioned time period. Due to the complex differences in player roles in football, we choose to fit separate models for each position, which gives us even less sample size for projecting player performance. As such, we choose to use Random Forest and XGBoost models, given their flexibility in capturing non-linear interactions between variables and interpretable feature importance scores, especially important when working with such a high number of predictor variables. These approaches strike a balance between flexibility and interpretability with less risk of overfitting when compared to more complex models such as Neural Networks, which is valuable due to the smaller sample of data.

Results

Model Choices / Feature Selection

For each position (quarterback/running back/wide receiver/tight end/offensive line/defensive line/edge rusher/linebacker/cornerback/safety), we fit separate Random Forest and XGBoost models, with different feature sets based on the player's role on the field. We choose

the final model based on which produces better out-of-sample results. Below we can see a list of the feature sets for each position. The response variable for all models is the player's PFF WAA in the following season.

Quarterback: snap counts, years in college football, high school rating/stars/ranking (from 247Sports), transfer portal stars (from 247Sports), conference, height/weight, WAA, pass completions/attempts/completion percentage/yards per attempt, passing yards/touchdowns/interceptions, rushing attempts/yards/touchdowns/yards per carry, longest rush, fumbles

Running Back/Fullback: snap counts, years in college football, high school rating/stars/ranking (from 247Sports), transfer portal stars (from 247Sports), conference, height/weight, WAA, rushing attempts/yards/touchdowns/yards per carry, longest rush, fumbles, receptions, receiving yards/touchdowns/yards per reception, longest reception, average speed (from 0-10 yards as well as 10-20 yards), maximum speed (0-10 yards, 10-20 yards, and overall), acceleration and deceleration (from 0-5 yards, 5-10 yards, 10-15 yards, 15-20 yards), position (to distinguish between running backs and fullbacks), total distance covered per game

Wide Receiver: snap counts, years in college football, high school rating/stars/ranking (from 247Sports), transfer portal stars (from 247Sports), conference, height/weight, WAA, rushing attempts/yards/touchdowns/yards per carry, longest rush, fumbles, receptions, receiving yards/touchdowns/yards per reception, longest reception, average speed (from 0-10 yards as well as 10-20 yards), maximum speed (0-10 yards, 10-20 yards, and overall), acceleration and deceleration (measured in 0.5 seconds, first second, first 1.5 seconds, first two seconds), total distance covered per game

Tight End: snap counts, years in college football, high school rating/stars/ranking (from 247Sports), transfer portal stars (from 247Sports), conference, height/weight, WAA, rushing attempts/yards/touchdowns/yards per carry, longest rush, fumbles, receptions, receiving yards/touchdowns/yards per reception, longest reception, average speed (from 0-10 yards as well as 10-20 yards), maximum speed (0-10 yards, 10-20 yards, and overall), acceleration and deceleration (measured in 0.5 seconds, first second, first 1.5 seconds, first two seconds), position (to distinguish between running backs and fullbacks), total distance covered per game, get-off time (first second, first 1.5 seconds, first 2 seconds)

Offensive Line: snap counts, years in college football, high school rating/stars/ranking (from 247Sports), transfer portal stars (from 247Sports), conference, height/weight, WAA, average speed (from 0-10 yards as well as 10-20 yards), maximum speed (0-10 yards, 10-20 yards, and overall), acceleration and deceleration (measured in 0.5 seconds, first second, first 1.5 seconds, first two seconds), total distance covered per game, get-off time (first second, first 1.5 seconds, first 2 seconds)

Cornerback: snap counts, years in college football, high school rating/stars/ranking (from 247Sports), transfer portal stars (from 247Sports), conference, height/weight, WAA, average speed (from 0-10 yards as well as 10-20 yards), maximum speed (0-10 yards, 10-20 yards, and overall), acceleration and deceleration (measured in 0.5 seconds, first second, first 1.5 seconds, first two seconds), total distance covered per game, closing speed on targets

Safety: snap counts, years in college football, high school rating/stars/ranking (from 247Sports), transfer portal stars (from 247Sports), conference, height/weight, WAA, tackles, tackles for loss, sacks, QB hurries, interceptions, interception yards, pass deflections, defensive touchdowns, average speed (from 0-10 yards as well as 10-20 yards), maximum speed (0-10 yards, 10-20

yards, and overall), acceleration and deceleration (measured in 0.5 seconds, first second, first 1.5 seconds, first two seconds), total distance covered per game, closing speed on targets

Linebacker: snap counts, years in college football, high school rating/stars/ranking (from 247Sports), transfer portal stars (from 247Sports), conference, height/weight, WAA, average speed (from 0-10 yards as well as 10-20 yards), maximum speed (0-10 yards, 10-20 yards, and overall), acceleration and deceleration (measured in 0.5 seconds, first second, first 1.5 seconds, first two seconds), total distance covered per game, closing speed on targets

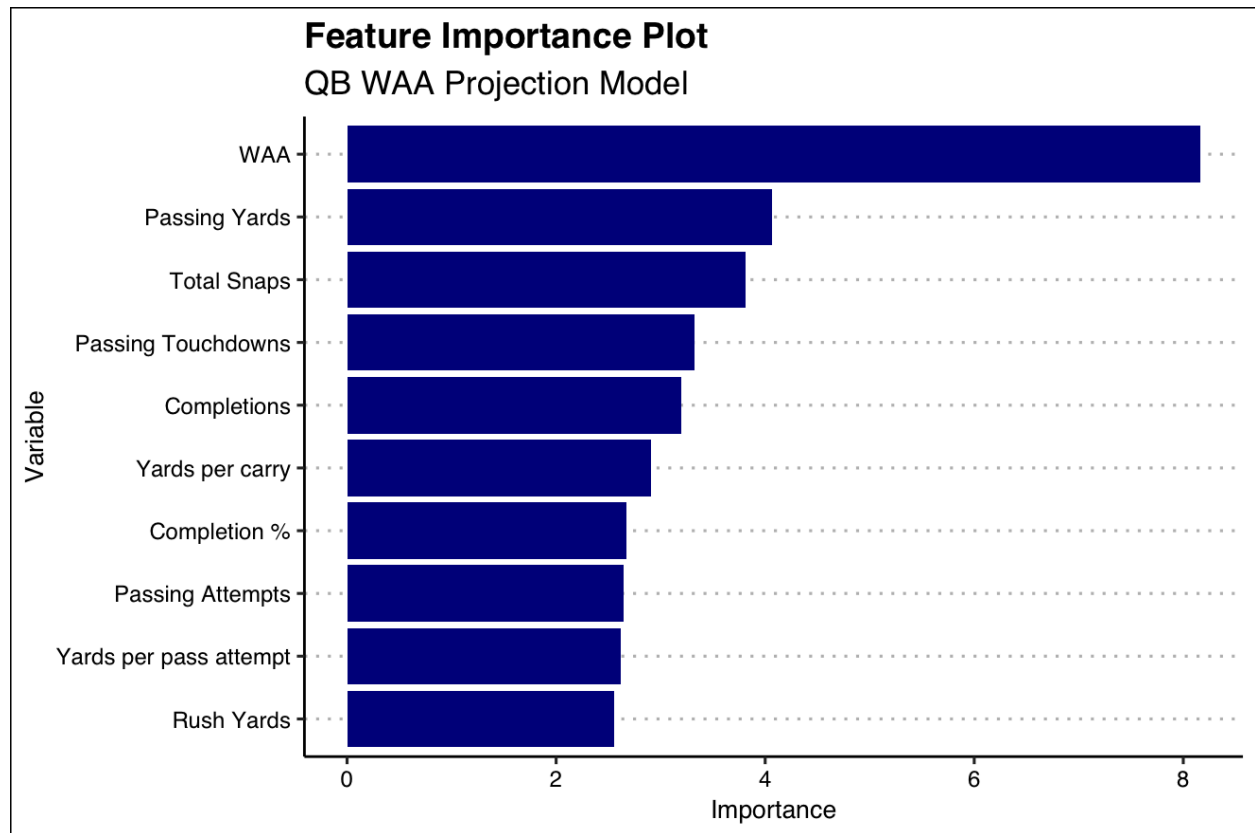
Edge Rusher: snap counts, years in college football, high school rating/stars/ranking (from 247Sports), transfer portal stars (from 247Sports), conference, height/weight, WAA, average speed (from 0-10 yards as well as 10-20 yards), maximum speed (0-10 yards, 10-20 yards, and overall), acceleration and deceleration (measured in 0.5 seconds, first second, first 1.5 seconds, first two seconds), total distance covered per game, closing speed on targets, get-off time (first second, first 1.5 seconds, first 2 seconds)

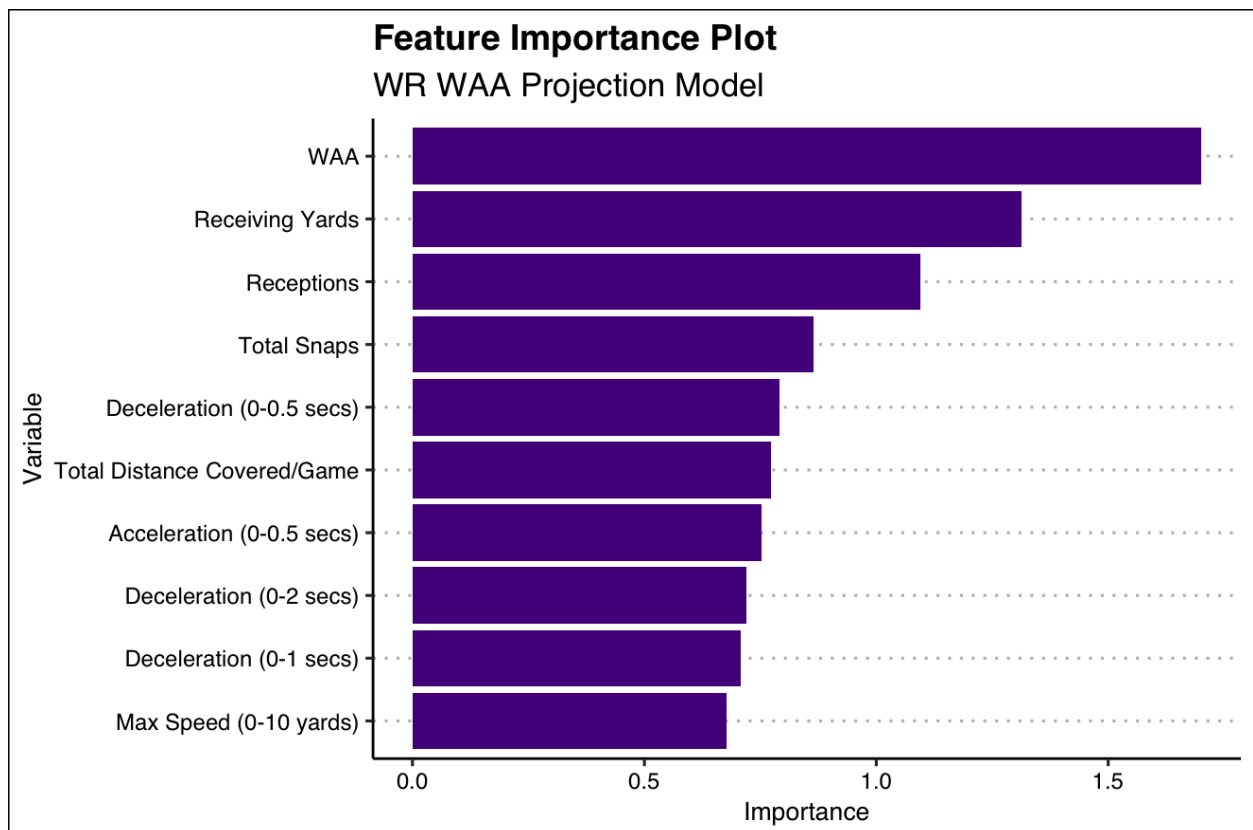
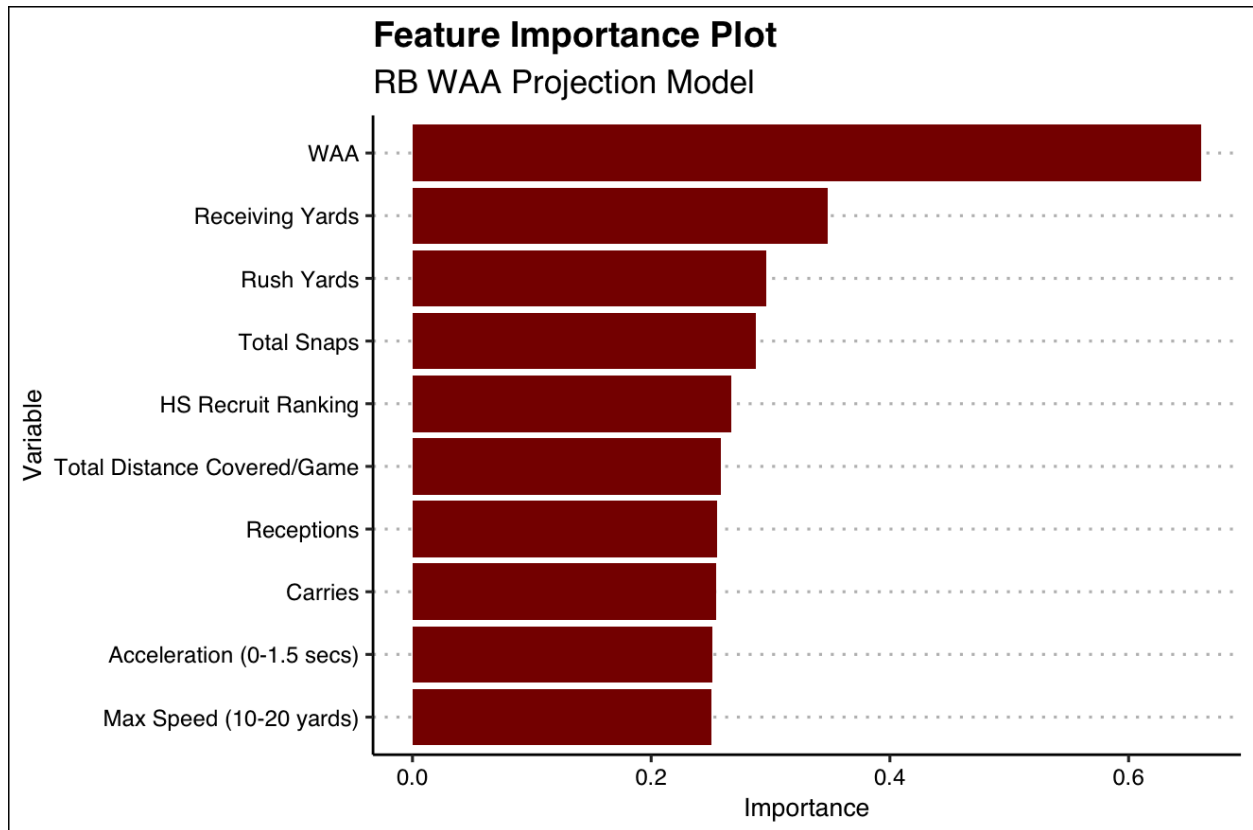
Defensive Line: snap counts, years in college football, high school rating/stars/ranking (from 247Sports), transfer portal stars (from 247Sports), conference, height/weight, WAA, tackles, tackles for loss, sacks, QB hurries, interceptions, interception yards, pass deflections, defensive touchdowns, average speed (from 0-10 yards as well as 10-20 yards), maximum speed (0-10 yards, 10-20 yards, and overall), acceleration and deceleration (measured in 0.5 seconds, first second, first 1.5 seconds, first two seconds), total distance covered per game, closing speed on targets, get-off time (first second, first 1.5 seconds, first 2 seconds)

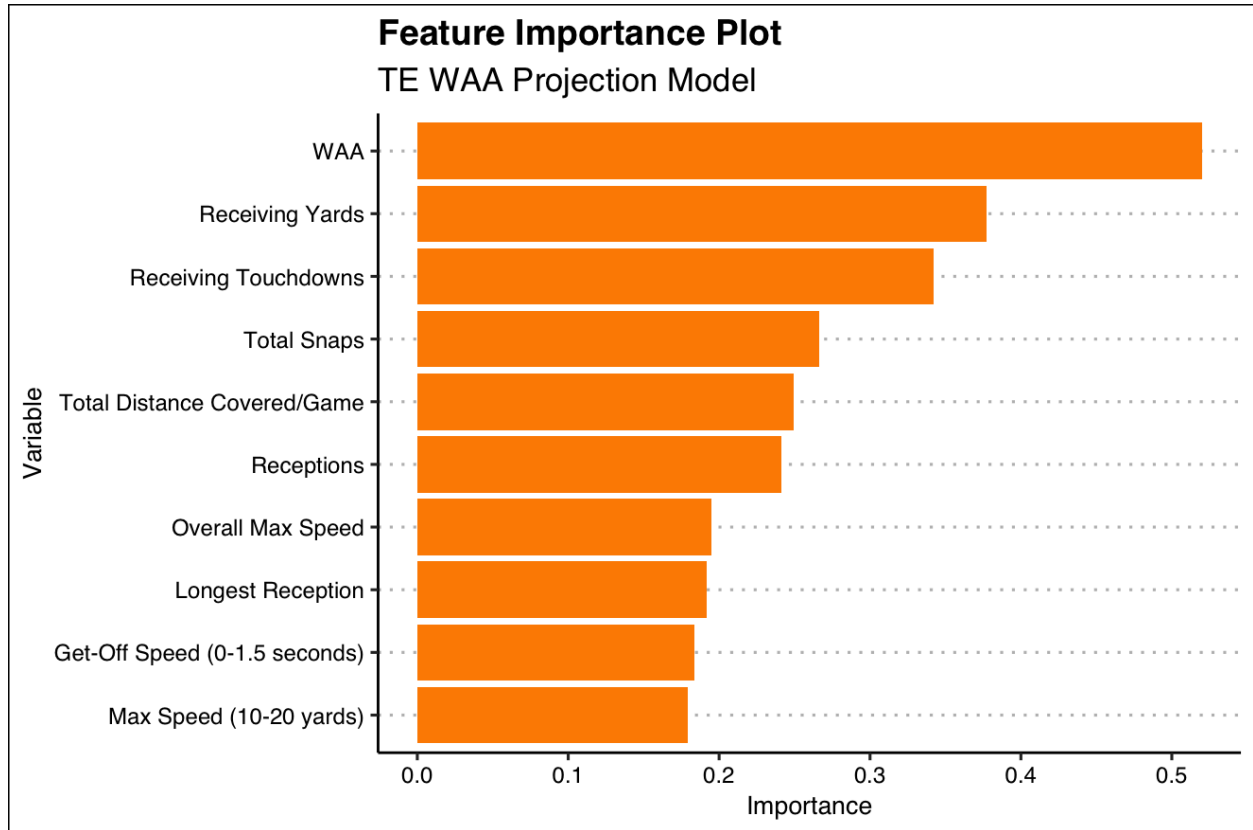
Feature Importance

The benefit of using Random Forests and XGBoosts for this prediction task, especially when comparing to the PFF approach, is the interpretability of the feature importance scores.

Through these, we can see what drives WAA in the next season and which variables are contributing the most to our predictions, for each position. Below are the feature importance plots for each skill position model:







To evaluate these models on out-of-sample data, we utilize leave-one-season-out cross-validation. Iterating over all seasons from 2021 to 2025, we alternate training the model on all seasons *except* one and then applying the model to the unseen data. We can then compare the model's performance in projecting WAA to the PFF WAA projections, weighted by snaps played so that players with higher samples (which we are more confident in our projections for) are emphasized. Below we can see the results for each position:

Position	R-Squared (PFF Projections)	R-Squared (New Model)
Quarterback	0.2501	0.1815 (XGBoost)
Running Back	0.1356	0.0823 (Random Forest)
Wide Receiver	0.1676	0.1502 (XGBoost)
Tight End	0.2118	0.1784 (XGBoost)

Offensive Line	0.2643	0.2514 (XGBoost)
Cornerback	0.0871	0.0897 (XGBoost)
Safety	0.0781	0.1096 (XGBoost)
Linebacker	0.0895	0.1033 (Random Forest)
Edge Rusher	0.2738	0.2592 (XGBoost)
Defensive Line	0.2568	0.2612 (Random Forest)
Overall	0.202	0.2009

One important note is that including box score statistics worsened the model for all defensive positions except safety and defensive line. This suggests there is a lot of noise in traditional defensive stats, and that athleticism metrics and other information have more predictive power in projecting future performance than just raw tackles, interceptions, etc. We see that our models were able to outperform PFF projections or come close much more on the defensive side than the offensive side. In particular, skill positions were the toughest to project accurately. With so much of offensive performance being scheme or quarterback dependent, it makes sense that defensive performance is more repeatable and predictable from season-to-season. Our model beat PFF's performance on four positions: cornerback, safety, linebacker, and defensive line. Overall, our models were barely worse than PFF projections at the player-season level. When summing up projected WAA by every team, we observe that PFF's projections have a better correlation with total team wins, with a 0.2453 R-Squared compared to our model's 0.2174 R-Squared. While our performance is similar at the individual level, there is still ground to make up when it comes to directly predicting winning impact.

2026 Results

When applying our model to current data, we can predict who the best players in college football will be next season. Below we see the top five projections for both all players and non-quarterbacks:

Highest Predicted WAA			
2026 CFB Season			
	PLAYER	POSITION	PREDICTED WAA
	Julian Sayin	QB	0.78
	Byrum Brown	QB	0.55
	Dante Moore	QB	0.55
	Gunner Stockton	QB	0.54
	Noah Fifita	QB	0.49

Highest Predicted WAA			
Non-Quarterbacks, 2026 CFB Season			
	PLAYER	POSITION	PREDICTED WAA
	KJ Bolden	S	0.29
	Jeremiah Smith	WR	0.26
	Dylan Labarbera	ED	0.22
	Jackson Harris	WR	0.22
	KJ Duff	WR	0.21

At the top, we see familiar signal-callers on P4 programs such as Julian Sayin (Ohio State), Dante Moore (Oregon), and Gunner Stockton (Georgia). When filtering for non-quarterbacks, known superstars such as KJ Bolden (Georgia) and Jeremiah Smith (Ohio State) stand out. We see that our metric seems to be properly valued in the transfer portal, as Byrum Brown (USF to Auburn) and Jackson Harris (Hawai'i to LSU) both got snatched up by P4 schools when the portal opened.

We can also use our models to project the best teams in the 2026 CFB season by summing up their roster's projected WAA. Below we see the results:

Highest Predicted WAA			
2026 CFB Rosters			
RANK	TEAM	PREDICTED WAA	LOGO
1	TEXAS TECH Red Raiders	2.09	
2	OREGON Ducks	1.96	
3	OHIO STATE Buckeyes	1.95	
4	TEXAS Longhorns	1.75	
5	GEORGIA Bulldogs	1.67	
6	NOTRE DAME Fighting Irish	1.61	
7	OLE MISS Rebels	1.55	
8	INDIANA Hoosiers	1.45	
9	LSU Tigers	1.20	
10	MIAMI Hurricanes	1.15	

At the top, we see big NIL spenders Texas Tech, Oregon, and Ohio State. Indiana looks to remain strong, and the rest of the list is full of familiar faces such as Georgia, Notre Dame, and Miami.

All individual player projections for the 2026 season (based off of 2025 information) can be viewed in an interactive Shiny App linked [here](#). Simply choose the position in question, and one can search for any player and see how he will perform according to our models.

Conclusion

Through this work, we find that player performance in the transfer portal era can be modeled with reasonable accuracy, but remains inherently noisy due to role, scheme, and context

dependence. Our model performance varies by position: offensive skill positions are less predictable, while defensive positions show stronger signal, especially when using athleticism metrics rather than traditional box score statistics. At the individual player level, models approach the performance of existing benchmarks, while at the team level, our aggregated projections still lag behind PFF's more established systems. The framework provides a scalable, position-agnostic tool for evaluating transfer portal players, helping programs make more informed decisions about roster construction and NIL allocation. Future work should focus on incorporating team context (scheme, coaching, surrounding talent) and improving offensive evaluation through better features.